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QUESTION 1: METHOD OVERRIDING

```
class Vehicle {  
    int baseSpeed = 60;  
  
    void maxSpeed() {  
        System.out.println("Maximum Speed of Vehicle: " + baseSpeed + " km/h");  
    }  
}  
  
class Car extends Vehicle {  
  
    @Override  
    void maxSpeed() {  
        System.out.println("Maximum Speed of Car: " + (baseSpeed + 60) + " km/h");  
    }  
}  
  
class Bike extends Vehicle {  
  
    @Override  
    void maxSpeed() {  
        System.out.println("Maximum Speed of Bike: " + (baseSpeed + 100) + " km/h");  
    }  
}
```

```
public class VehicleTrackingSystem {  
    public static void main(String[] args) {  
  
        Vehicle v = new Vehicle();  
        Car c = new Car();  
        Bike b = new Bike();  
  
        v.maxSpeed();  
        c.maxSpeed();  
        b.maxSpeed();  
    }  
}
```

QUESTION 2: METHOD OVERLOADING

```
class Payment {  
  
    void pay(String cardNumber, String cardHolder, double amount) {  
        System.out.println("Payment through Credit Card");  
        System.out.println("Card Number: " + cardNumber);  
        System.out.println("Card Holder: " + cardHolder);  
        System.out.println("Amount Paid: ₹" + amount);  
    }  
  
    void pay(long accountNumber, String ifscCode, double amount) {  
        System.out.println("\nPayment through Bank Account");  
        System.out.println("Account Number: " + accountNumber);  
    }  
}
```

```
        System.out.println("IFSC Code: " + ifscCode);
        System.out.println("Amount Paid: ₹" + amount);
    }
```

```
void pay(String upid, double amount) {
    System.out.println("\nPayment through UPI");
    System.out.println("UPI ID: " + upid);
    System.out.println("Amount Paid: ₹" + amount);
}
}
```

```
public class PaymentProcessingSystem {
    public static void main(String[] args) {

        Payment p = new Payment();

        p.pay("1234-5678-9012-3456", "Atharv", 5000.0);

        p.pay(123456789012L, "SBIN0001234", 2500.0);

        p.pay("atharv@upi221", 1000.0);
    }
}
```